

SPECIAL TESTS — LUMBO-SACRAL REGION

I. TEST FOR LUMBAR INSTABILITY

1. PASSIVE LUMBAR EXTENSION TEST

PURPOSE

To assess lumbar spine instability.

PROCEDURE

- Patient lies prone.
- Examiner lifts both the patient's legs together by holding around the ankles.
- The legs are elevated while the knees remain extended.
- Observe the patient's symptoms and lumbar response.

POSITIVE SIGN

- Development/reproduction of low-back pain
- Feeling of heaviness or instability in the lumbar region.

INDICATES

Lumbar segmental instability.

II. TESTS FOR MUSCLE DYSFUNCTION

2. 90–90 STRAIGHT LEG RAISING TEST

PURPOSE

To assess hamstring muscle tightness/flexibility.

PROCEDURE

- Patient lies supine.
- Flex the hip and knee to approximately 90°.
- Stabilize the thigh.
- Slowly extend the knee while maintaining hip flexion.

POSITIVE SIGN

- Inability to achieve full knee extension.
- Increased resistance/tightness in the posterior thigh.

INDICATES

Hamstring tightness/shortness.

3. TRENDELENBURG TEST

PURPOSE

To assess hip abductor muscle function, especially gluteus medius.

PROCEDURE

- Patient stands on one leg.
- Ask the patient to lift the opposite foot from the floor.
- Observe the position of the pelvis.

POSITIVE SIGN

- Pelvis drops on the side of the lifted/non-weight-bearing leg.

INDICATES

Weakness of the hip abductors of the weight-bearing leg, particularly gluteus medius.

III. TESTS FOR NEUROLOGICAL DYSFUNCTION

4. STRAIGHT LEG RAISING TEST (SLR/LASÈGUE'S TEST)

PURPOSE

To assess lumbosacral nerve-root irritation, particularly due to PIVD.

PROCEDURE

- Patient lies supine.
- Keep the knee fully extended.
- Slowly raise the patient's straight leg by flexing the hip.
- Note the angle and symptoms produced.

POSITIVE SIGN

- Reproduction of radicular/sciatic pain, typically between about 30–70° of hip flexion.

INDICATES

Lumbosacral nerve-root irritation/compression, commonly associated with PIVD.

5. SLUMP TEST

PURPOSE

To assess neural tissue mechanosensitivity, particularly lumbosacral nerve-root involvement.

PROCEDURE

Patient sits with:

1. Hands behind back.
2. Thoracic/lumbar spine flexed.
3. Cervical spine flexed.
4. One knee is extended.
5. Ankle is dorsiflexed.

The examiner progressively adds these components while observing symptoms.

POSITIVE SIGN

- Reproduction of the patient's familiar neural/radicular symptoms.
- Symptoms change with release of one component, particularly cervical extension or ankle movement.

INDICATES

Neural tissue/nerve-root involvement, such as lumbar radiculopathy.

IV. TESTS FOR JOINT DYSFUNCTION

6. SCHOBER'S TEST

PURPOSE

To assess lumbar spine flexion/mobility.

PROCEDURE

- Patient stands erect.
- Mark the lumbosacral region and a point 10 cm above it.
- Ask the patient to bend forward maximally.
- Measure the distance between the marks again.

POSITIVE SIGN

Insufficient increase in distance during forward flexion.

INDICATES

Reduced lumbar flexion/mobility, commonly seen in conditions such as ankylosing spondylitis.

7. FABER / PATRICK'S TEST

FABER = Flexion – ABduction – External Rotation

PURPOSE

To assess hip and sacroiliac joint pathology.

PROCEDURE

- Patient lies supine.
- Place the tested leg in:
 - Hip flexion
 - Abduction
 - External rotation
- Place the lateral ankle over the opposite thigh.
- Stabilize the opposite pelvis.
- Gently press the flexed knee downward.

POSITIVE SIGN

- Pain reproduced in the hip/groin → suggests hip pathology.
- Pain in the sacroiliac region → may indicate SI joint pathology.
- Restricted ROM may also be significant.

INDICATES

Hip or sacroiliac joint dysfunction.

8. GAENSLEN'S TEST

PURPOSE

To assess sacroiliac joint dysfunction/pathology.

PROCEDURE

- Patient lies near the edge of the table.
- One hip and knee are maximally flexed toward the chest.
- The opposite leg is allowed to hang off the edge of the table.
- Apply gentle pressure to produce opposite movements at the pelvis.

POSITIVE SIGN

- Reproduction of pain in the sacroiliac region.

INDICATES

Sacroiliac joint dysfunction.

9. GILLET'S TEST

PURPOSE

To assess sacroiliac joint mobility.

PROCEDURE

- Patient stands.
- Examiner palpates the PSIS and sacrum.
- Patient flexes one hip and knee, lifting the foot from the floor.
- Observe movement of the PSIS relative to the sacrum.

POSITIVE SIGN

- Abnormal or restricted movement of the PSIS compared with the opposite side.

INDICATES

Sacroiliac joint hypomobility/dysfunction.

10. MARCH'S TEST

PURPOSE

To assess sacroiliac joint movement/dysfunction.

PROCEDURE

- Patient stands.
- Examiner palpates the PSIS.
- Patient marches by lifting one knee at a time.
- Observe the movement of the PSIS.

POSITIVE SIGN

- Abnormal/asymmetrical movement of the PSIS.

INDICATES

Sacroiliac joint dysfunction or altered mobility.

★ REVISION TABLE — LUMBO-SACRAL SPECIAL TESTS

Category	Test	Indicates
Lumbar instability	Passive Lumbar Extension Test	Lumbar instability
Muscle dysfunction	90–90 SLR	Hamstring tightness
Muscle dysfunction	Trendelenburg Test	Hip abductor weakness
Neurological dysfunction	SLR	Lumbosacral nerve-root irritation
Neurological dysfunction	Slump Test	Neural mechanosensitivity/root involvement
Joint dysfunction	Schober’s Test	Lumbar mobility
Joint dysfunction	FABER/Patrick’s	Hip/SI pathology
Joint dysfunction	Gaenslen’s	SI joint dysfunction
Joint dysfunction	Gillet’s	SI joint mobility
Joint dysfunction	March’s Test	SI joint dysfunction